

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Evaluate $\int_0^{2\sqrt{4-y^2}} \int_0^{\sqrt{4-y^2}} \cos(x^2 + y^2) dx dy$ by changing into polar integration. 5
 - b) Evaluate: $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$ 5
 - c) Find the volume in the first octant bounded by the coordinate planes, the cylinder $x^2 + y^2 = 4$ and $z + y = 3$. 5
 2. a) Solve the differential equation $y'' + (1 - x^2)y = 0$ by using power series method. 7
 - b) Define Legendre's equation. Also derive the solution of Legendre's equation. 8
- OR**
- If $J_n(x)$ represents the Bessel's function of order n then show that:
- i. $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$.
 - ii. $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$.
3. a) State First Shifting Theorem of Laplace transforms. 8
 - i. Find the Laplace transform of $te^{-t} \cosh t$
 - ii. Find the inverse Laplace transform of $\frac{1}{s^2 - 5s + 6}$
 - b) Solve the initial value problem:
 $y'' + 4y' + 3y = e^{-t}, y(0) = y'(0) = 1$ by using Laplace transform. 7
 4. a) If $\Phi = \log(x^2 + y^2 + z^2)$ then find $\text{div}(\text{grad} \Phi)$. 5

- b) Find the directional derivative of $f = x^2yz + 4xz^2$ at $(1, -2, 1)$ in the direction $2\vec{i} - \vec{j} - 2\vec{k}$. 5
 - c) Calculate: $\oint_C \vec{F} \cdot d\vec{r}$ if $\vec{F} = (y, z, x), C: \vec{r} = (t, t^2, t^3)$, from $(0, 0, 0)$ and $(2, 4, 8)$. 5
 5. a) Define Green's Theorem. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (\sin y, \cos x)$ and c is the triangle with vertices $(0, 0), (\pi, 0), (\pi, 1)$, by using Green's theorem. 8
 - b) Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by using Stoke's theorem, where $\vec{F} = (y^2, z^2, x^2)$ and c is the boundary of the surface $S: x + y + z = 1$ in the first octant. 7
- OR**
- Evaluate the surface integral $\iint_S \vec{F} \cdot \vec{n} dA$ where $\vec{F} = (x^2, y^2, z^2)$ and $s: \vec{r} = (u \cos v, u \sin v, 3v), 0 \leq u \leq 1, 0 \leq v \leq 2\pi$
6. a) Find the fourier series of $f(x) = x + |x|$ for $-\pi < x < \pi$. 7
 - b) Expand the function $f(x) = x^2$ in the interval $0 < x < \pi$ in half range Fourier cosine series and show that $\frac{\pi^2}{6} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$ 8
 7. Solve Any Two: 2x5
 - a) Find the general solution of $2u_x + 2u_y - u = 0$.
 - b) Derive the one-dimensional traffic flow model using conservation law.
 - c) Show that the value under integral sign $\int_{(4,0,3)}^{(-1,1,2)} [(yz + 1)dx + (xz + 1)dy + (xy + 1)dz]$ is exact and evaluate the integral.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE

Course: Microprocessor and ALP (New)

Semester: Fall

Year : 2023

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define microprocessor. Differentiate between Von Newmann and Harvard architecture along with its block diagram. 7
b) Draw the functional block diagram of intel 8085 microprocessor and explain each block. 8
2. a) Draw and explain a well-labelled timing diagram of the instruction LXI B, D050H and calculate the time required to execute the instruction if frequency = 4Mhz. 8
b) Write an ALP in 8085 to transfer 10 bytes of data of memory address starting from 5000H to 6000H. Comment in each line indicate full marks. 7
3. a) What are the major changes and advantages of 8086 over 8085 microprocessors? Explain addressing modes of 8086. 8
OR
What are the types of FLAGS available in 8086 microprocessors? Show the Flag position and explain in brief with examples. 7
b) Write a 8086 program to find square root of a number.
4. a) What are the assembler directives of 8086? List out some common directives that are used while programming. 7
OR
What are Macros? Explain DOS interrupt functions.
b) Design an address decoding circuit to interface 2KB RAM, 4 KB ROM and 8 KB RAM with initial address 2000H. 8
5. a) Define parallel interface. Explain 8255 PPI. 7
b) Draw and explain the architecture of 8251 USART along with its pins. 8
6. a) What is the difference between polled and Vectored interrupt handling technique? Explain with necessary diagram. 7
b) What are the sources of interrupt? Explain interrupt vector Table. 8

7. Write short notes on: (Any two)

- a) Microcontroller and applications
- b) Synchronous and Asynchronous Bus
- c) 8259 PIC

2×5

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Operating System (New)

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is operating system? Explain OS as resource manager and an extended machine. 8
- b) Producer Consumer problem can be solved with the help of various techniques. Suggest any one of the techniques to solve producer consumer with detailed explanation. 7
2. a) Given the following set of information, calculate the average waiting time and average turn-around time using FCFS, SJF, RR (Quantum = 2) and HRRN. 8

Process	Arrival Time	Service Time (Burst Time)
A	0	7
B	2	2
C	3	3
D	5	8

OR

- Define threads. Explain the multithreading model in detail.
- b) Define PCB. Explain the Readers and Writers problem with necessary pseudocode. 7
3. a) Consider the following page reference strings: 2, 3, 4, 5, 3, 2, 6, 9, 3, 2, 3, 4, 8, 9, 4, 3, 2, 3, 4, 9. How many page faults would occur for each of the following page replacement algorithms assuming 3 pages a frame? In each case calculate fault ratio. 8
 - i. FIFO page replacement
 - ii. Optimal page replacement
 - iii. LRU page replacement

- b) Define Thrashing. Explain the concept of paging with a suitable example. 7

OR

What is difference between Paging and Segmentation? What are different page-table structures? Explain any one.

4. a) Why is deadlock state more critical than starvation? Describe resource allocation graph with a deadlock, with a cycle but no deadlock. 7
- b) What are different ways to input output? Explain the principles of I/O software. 8
5. a) What is file? With help of necessary diagram, explain file system allocation techniques. 8
- b) Explain the concept, characteristics and role of Distributed Operating System. 7
6. a) Define pre-paging. Explain WS page replacement algorithm with example. 8
- b) Explain about Memory wall and bottleneck problem for Operating System with necessary diagrams. 7
7. Write short notes on: (Any two) 2×5
 - a) Relocation and protection
 - b) RAID
 - c) Context switching

POKHARA UNIVERSITY

Semester: Fall

Year : 2023

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

Bachelor
Programme: BE
Course: Database Management System (New)

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is data abstraction in DBMS? Explain in detail. 7
b) Consider you are asked to design a database for the Exam section of your college. Draw its ER diagram assuming required entities and their attributes. 8
2. a) Convert the ER diagram that you designed in question no 1 b) into relational schema. 7
b) What are the views? Consider the table **tbl_emp** as follows: 8

Emp_id	Emp_name	Salary	Department	Date_of_joining
101	Anish	20000	Packing	2070-02-01
102	Rojina	18000	Cleaning	2075-04-06
103	Sita	35000	Polishing	2078-09-12

Write the SQL statements for the following:

- i. Create the above table by considering Emp_id as primary key and insert the above records.
- ii. Change the Department of Anish to marketing.
- iii. Increase the salary of employees whose department is Cleaning by 12%.
- iv. Find the name of employees having salary greater than 16000 and who joined after 2072-11-25
- v. Add a new column Address to the above table.
- vi. Delete the entire table.

OR

Why is joining in SQL necessary? Explain Inner Join, Natural Join and Outer Join with suitable examples.

3. a) What are the different types of integrity constraints? Explain with examples. 8

- b) What is denormalization? Why is it necessary? Explain in detail. 7
4. a) What is the role of views in security? Explain the concept of authorization with suitable examples. 7
- b) Explain the basic steps in query processing in detail. 8
5. a) Explain how the records are organized using fixed-length and variable-length records. 7
b) What is serializability and why is it needed? Explain the ACID properties in brief. 8
6. a) What is crash recovery? Explain log-based recovery method with example. 7

OR

- What is transaction rollback? Explain how the Remote Backup System provides high availability and recovery facility.
- b) List out the different categories of NoSQL databases. Explain the concept of blockchain with its properties. 8
7. Write short notes on: (Any two) 2×5
a) Nested Queries
b) Third Normal Form (3NF)
c) Lock-based protocols for concurrency control

Level: Bachelor
Programme: BE
Course: Computer Graphics (new)

POKHARA UNIVERSITY

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
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Attempt all the questions.

1. a) What is Computer Graphics? Differentiate between raster and random display with its architecture. 5 8
b) Why do we need hardware acceleration for graphics processing? Explain the significance of GPU in modern computer organization. Compare CPU with GPU. 5 7
2. a) What is scan conversion? How can equations be derived for drawing lines with slope greater than one using Bresenham's line drawing algorithm? Explain. 6 8
b) What is composite transformation. Explain why do we need homogeneous coordinate system for transformation of computation in computer graphics. 2 7
3. a) How symmetric property of circle can be implemented to digitize circle having equation $(x-4)^2 + (y+5)^2 = 36$. Illustrate with necessary calculations using mid-point circle drawing algorithm. 8

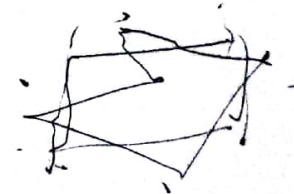
OR

- a) What is window and viewport? Magnify the triangle with vertices A(0,0) B(1,1) and C(5,2) to thrice its size while keeping C(5,2) fixed. 5 7
b) Explain the Sutherland Hodgeman polygon clipping algorithm considering the four different cases with example.

OR

- Let R be the rectangular window whose lower left hand corner is at L (-3, 1) and upper right-hand corner is at R (2, 6). Use Cohen - Sutherland algorithm to clip the line segments A (-4, 2) and B (-1, 7).
4. a) How are Two Dimensional Rotations different from Three 7

- Dimensional Rotations? Explain with the matrix representations. 5
Derive the blending function for parametric cubic curve using Hermite interpolation technique. 6
What are the drawbacks of Flat shading and Gouraud shading? How does normal vector interpolation scheme remove these drawbacks? 7
Explain painter's algorithm for visible surface detection with its necessary test cases. 8
Define GKS and PHIGS. Mention different types of Graphics file format used in computer graphics. 5
How OpenGL APIs can be used to maintain graphical standards? Explain. 7
7. Write short notes on: (Any two) 2x5
a) Color Models 6
b) 2D viewing pipeline 6
c) Frame Buffer



$$T(S, 2) \quad S(B, 8) \quad T(S, -2) \cdot P$$
$$T_4 \quad (-2, 2) \quad (5, 2)$$

Level: Bachelor
Programme: BE
Course: Data Communication (New)

POKHARA UNIVERSITY

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Simplex, Half Duplex and Full Duplex mode of data transmission. Briefly explain digital communication system with the help of general block diagram. 8
 - b) Differentiate between different types of transmission modes. Briefly Explain RS-232C. 7
 2. a) Define Signal. Briefly explain periodic and non-periodic signals, energy and power signals, deterministic and random signals, continuous and discrete time signals. 7
 - b) Define unit step signal with necessary mathematical expression and graphical illustration. Check the following system for linearity, causality, static & dynamic and time invariance. 8
$$y(n) = Ax(n) + B$$
 3. a) Compare and contrast between OSI and TCP/IP with necessary figures. 7
 - b) Define guided and unguided transmission media. State the advantages of optical fiber over other guided media. 8
 4. a) Why burst error is difficult to handle than single error? Explain why CRC is better than parity method of error detection. 7
- OR**
- Explain various transmission impairments of a communication system. 8
- b) Compare and contrast between lossless and lossy compression. 8
 5. a) Explain the importance of flow control. Briefly explain sliding window flow control mechanism. 7
 - b) Explain the multiplexing technique applied in digital telephony with their significance, application and multiplexing hierarchy. 8
 6. a) Is switching and routing same? Differentiate between message switching, circuit switching and packet switching. 8

- b) Differentiate between PSK, ASK and FSK modulation techniques with necessary diagram. 7

OR

Encode the Bit Stream 011010001 using the following scheme.

- i. NRZ - I
- ii. NRZ - L
- iii. Manchester
- iv. Differential Manchester

2×5

7. Write short notes on: (Any two)

- a) Cellular Telephony
- b) Point to Point protocol
- c) QPSK